

Matrix Inverse

Geometric Algorithms
Lecture 10

Practice Problem

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_1 - x_2 \\ 4x_2 \\ 3x_1 - 3x_2 \end{bmatrix}$$

Determine if the above transformation is onto, one-to-one, both, or neither

Answer

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_1 - x_2 \\ 4x_2 \\ 3x_1 - 3x_2 \end{bmatrix}$$

Objectives

» Motivate and define matrix inverses

Keywords

Matrix Transpose

Inner Product

Matrix Power

Square Matrix

Matrix Inverse

Invertible Transformation

1-1 Correspondence

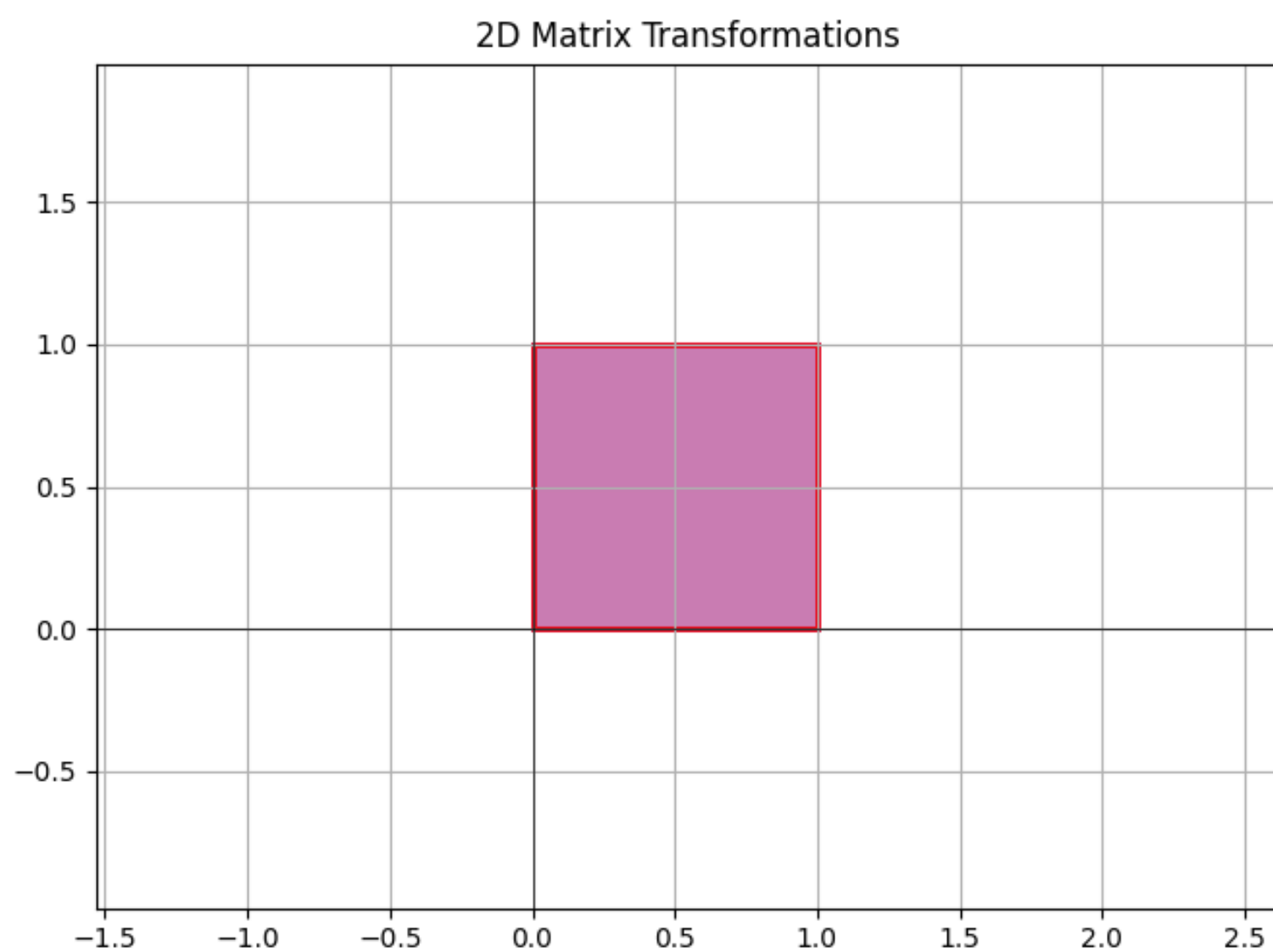
`numpy.linalg.inv`

Determinant

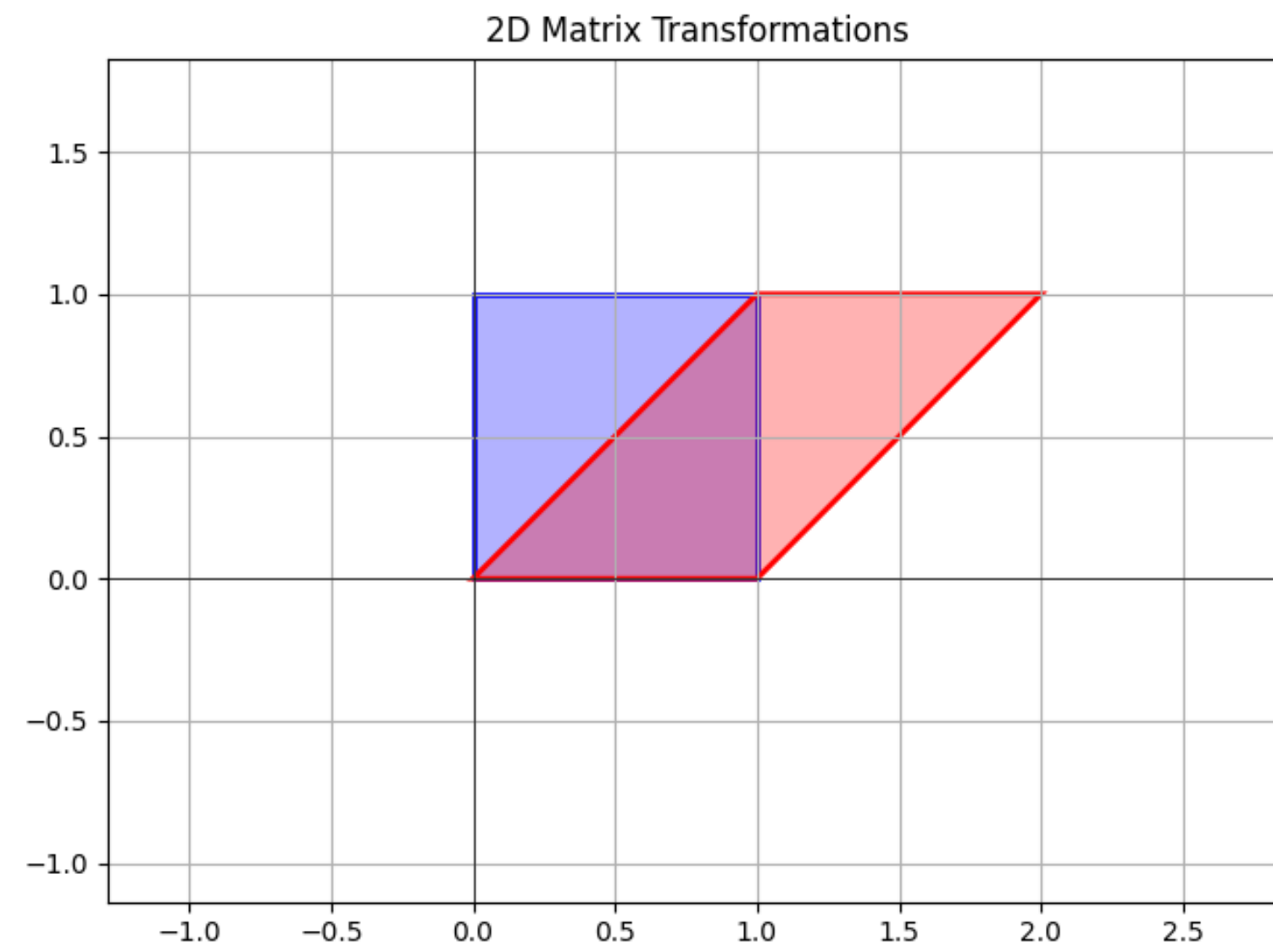
Invertible Matrix Theorem

Recap

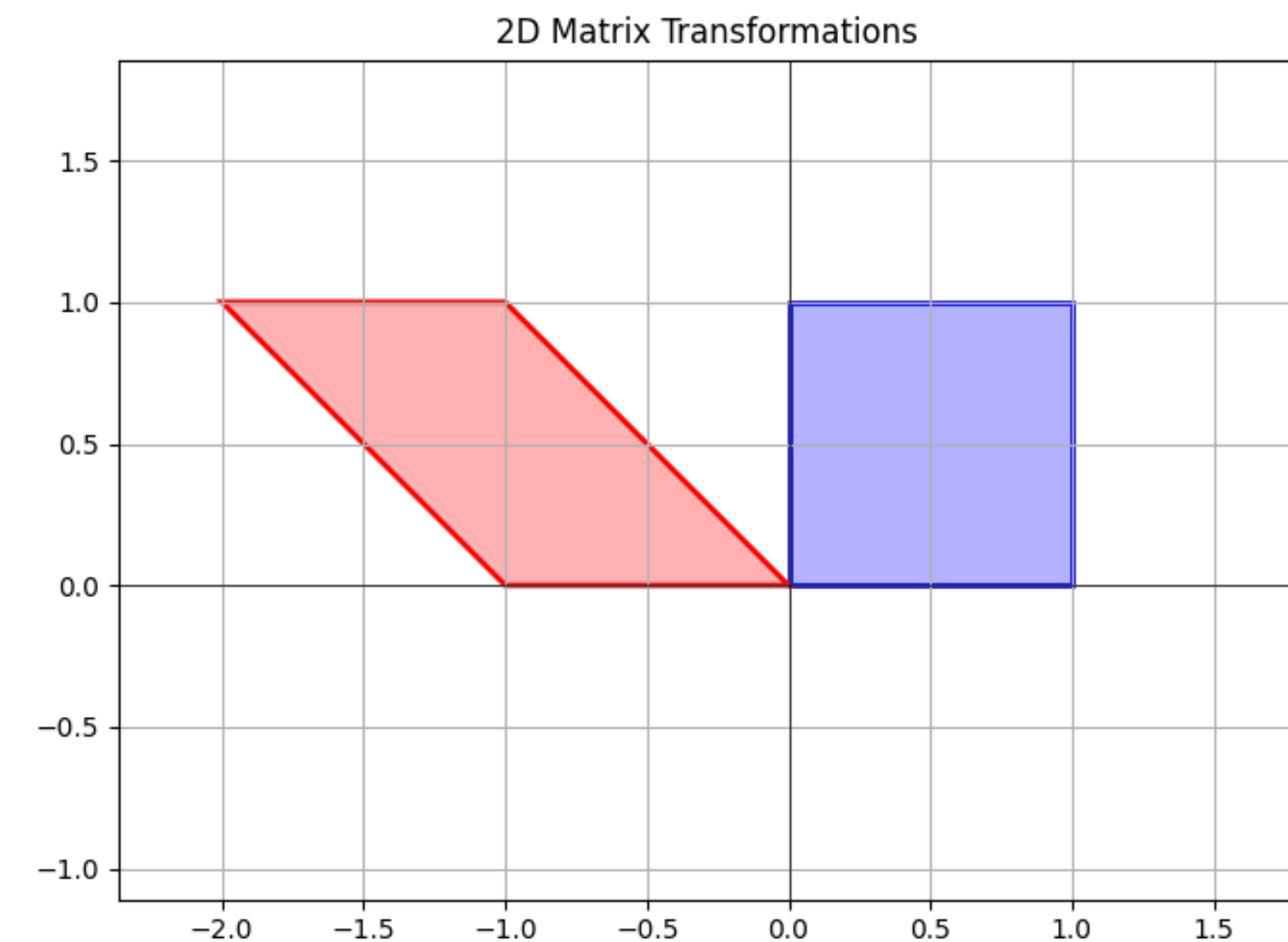
Recap: Shearing and Reflecting (Geometrically)



shear



reflect



Recap: The Key Fact

Thm. The composition of two linear transformations is a linear transformation

This means the composition of two matrix transformations can be implemented by a *single* matrix

Matrix Multiplication

Df. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ with columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$ the product $AB \in \mathbb{R}^{m \times p}$ is given by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix}$$

Replace each column of B with A multiplied by that column

Recap: Matrix Multiplication and Composition

Thm. For any matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ and any vector $\mathbf{v} \in \mathbb{R}^p$,

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

The matrix implementing the composition is the product of the two underlying matrices

Recap: Matrix "Interface"

multiplication just given

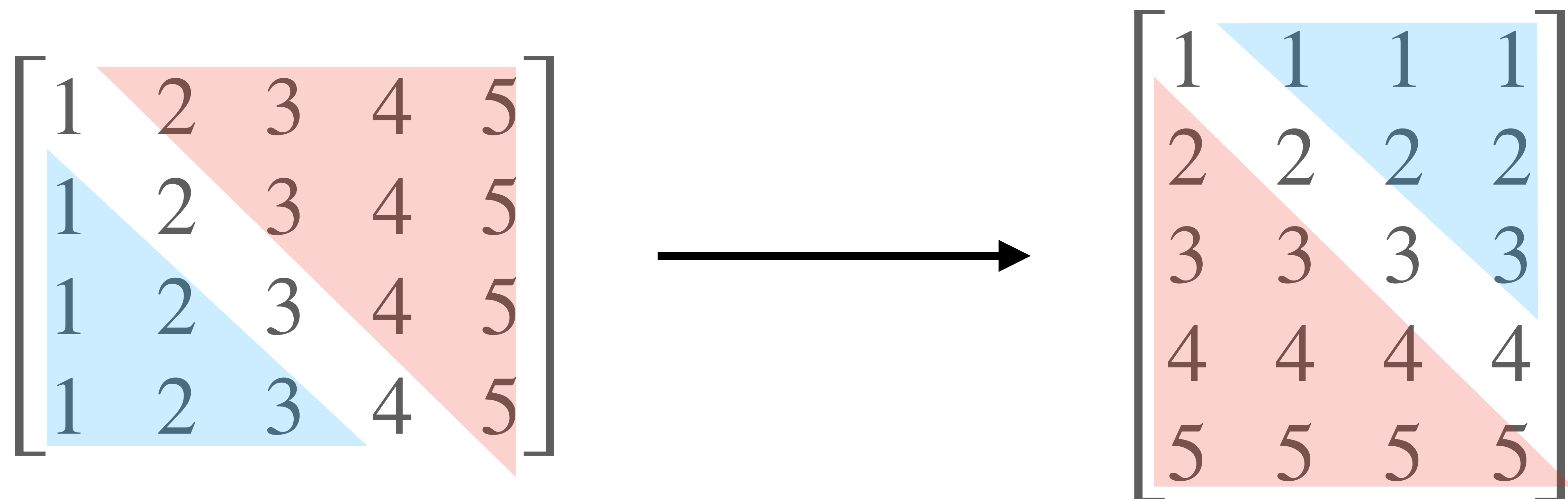
addition add entry-wise

scaling scale entry-wise

Recall: Transpose (Pictorially)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \end{bmatrix}$$

Recall: Transpose (Pictorially)



Recall: Transpose

Df. For $A \in \mathbb{R}^{m \times n}$ the **transpose** of A , written A^T , is the $n \times m$ matrix such that

$$(A^T)_{ij} = A_{ji}$$

Ex.

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Recall: Algebraic Properties

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T$$

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$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T \text{ Important: the order reverses!}$$

Transposes and Inner Products

For $\mathbf{v} \in \mathbb{R}^n$ note that $\mathbf{v}^T \in \mathbb{R}^{1 \times n}$ is a *single-row matrix*, sometimes called a **row vector**

For two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, is $\mathbf{u}^T \mathbf{v}$ defined?

Transposes and Inner Products

$$[u_1 \quad u_2 \quad u_3 \quad u_4] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

Transposes and Inner Products

$$[u_1 \quad u_2 \quad u_3 \quad u_4] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

Df. The **inner product** of two vectors **u** and **v** in $\mathbb{R}^n$ is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$

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Df. For $A \in \mathbb{R}^{n \times n}$ we write A^k for the k -fold product of A with itself

What should A^0 be? (we want $A^0 A^k = A^{0+k} = A^k$)

Matrix Powers (Computationally)

Use `numpy.linalg.matrix_power`

This can be *much* faster than doing a sequence of matrix multiplications (*why?*)

Recall: Final Warnings

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1. AB isn't necessarily equal to BA , even if both are well-defined

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2. If $AB = AC$ then it isn't necessary that $B = C$ even if $A \neq 0$
3. If $AB = 0$ (the zero matrix) it isn't necessary that $A = 0$ or $B = 0$

Matrix Operations (So Far)

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transpose

$$A^T$$

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$$A + B$$

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multiplication (powers)

$$AB$$

$$A^k$$

Matrix Inverses

Recall: The Identity Matrix

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Df. The $n \times n$ **identity matrix** is the matrix whose *diagonal* contains all 1s, and all other entries are 0s

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

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$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

Ex.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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The identity matrix implements the "do nothing" transformation. For any $\mathbf{v}$,

$$I\mathbf{v} = \mathbf{v}$$

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$$IA = AI = A$$

These may be different sizes

Recall: The Identity Matrix

$$\begin{array}{ccccc} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} & = & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \\ 2 \times 2 & 2 \times 4 & & 2 \times 4 & 4 \times 4 & & 2 \times 4 \end{array}$$

Basic Algebra

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Divide on both sides by 2 to get $x = 5$

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$\frac{1}{2}$ is the **reciprocal** or **multiplicative inverse** of 2

Wouldn't it be nice...

$$\mathbf{Ax} = \mathbf{b}$$

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A^{-1} is the **multiplicative inverse** of A

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$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$$

How do we solve this equation?

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Do all matrices have
inverses?

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inverses?

*No. If they did, then every linear
system would have a solution*

When does a matrix have
an inverse?

Square Matrices

Df. A matrix $A \in \mathbb{R}^{m \times n}$ is **square** if $m = n$

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

i.e., it has same number of rows as columns

Square Matrices are Special

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» *that can have inverses*

Matrix Inverses

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Df. For a matrix $A \in \mathbb{R}^{n \times n}$, an **inverse** of A is a matrix $B \in \mathbb{R}^{n \times n}$ such that

$$AB = I_n \text{ and } BA = I_n$$

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A is **invertible** if it has an inverse. Otherwise it is **singular**

Example

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Example

Reflection across the x -axis is it's own inverse

Non-Example

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Inverses are Unique

Thm. If B and C are inverses of A , then $B = C$

Solutions for Invertible Matrix Equations

Thm. For matrix $A \in \mathbb{R}^{n \times n}$, if A is invertible then the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$

Unique Solutions

If $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$, then it has

» exactly one solution for any choice of $\mathbf{b}$

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If $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$, then it has

» at least one solution for any choice of $\mathbf{b}$

» at most one solution for any choice of $\mathbf{b}$

Unique Solutions

If $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$, then it has

» T is onto

» T is one-to-one

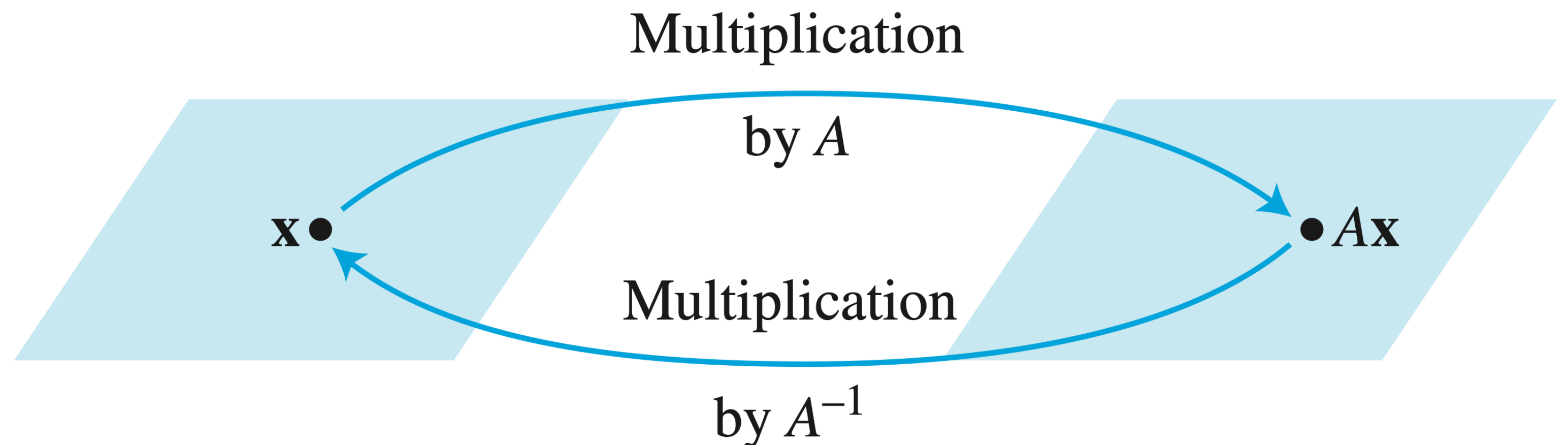
where T is implemented by A

Connection to Transformations

Df. A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **invertible** if there is a linear transformation S such that

$$S(T(\mathbf{v})) = \mathbf{v} \quad \text{and} \quad T(S(\mathbf{v})) = \mathbf{v}$$

for any $\mathbf{v}$ in $\mathbb{R}^n$



Connection to Transformations

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Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

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A matrix is invertible if it's possible to "undo" its transformation without "losing information"

Non-Example

Projection onto the x -axis in $\mathbb{R}^2$

Connection to Transformations

Connection to Transformations

Df. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **one-to-one correspondence** (bijection) if any vector $\mathbf{b}$ in $\mathbb{R}^n$ is the **image of exactly one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

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A transformation is a 1-1 correspondence if it is 1-1 and onto

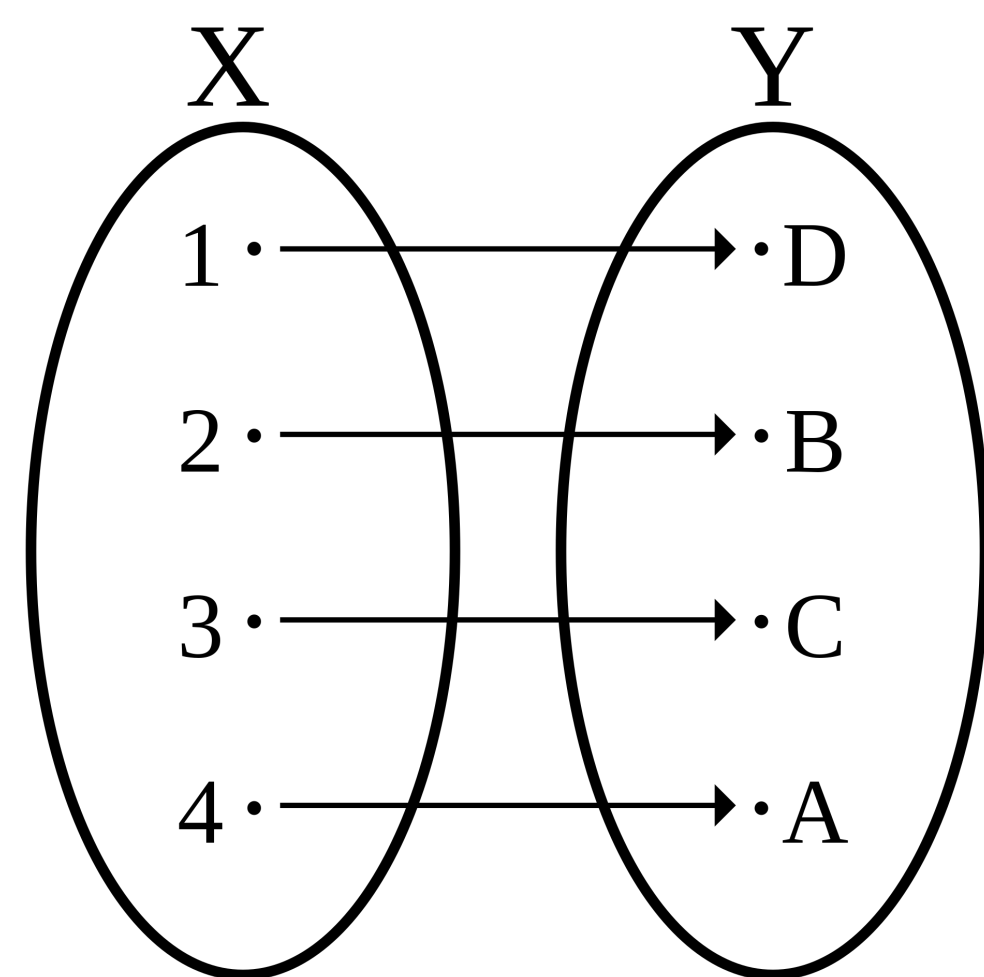
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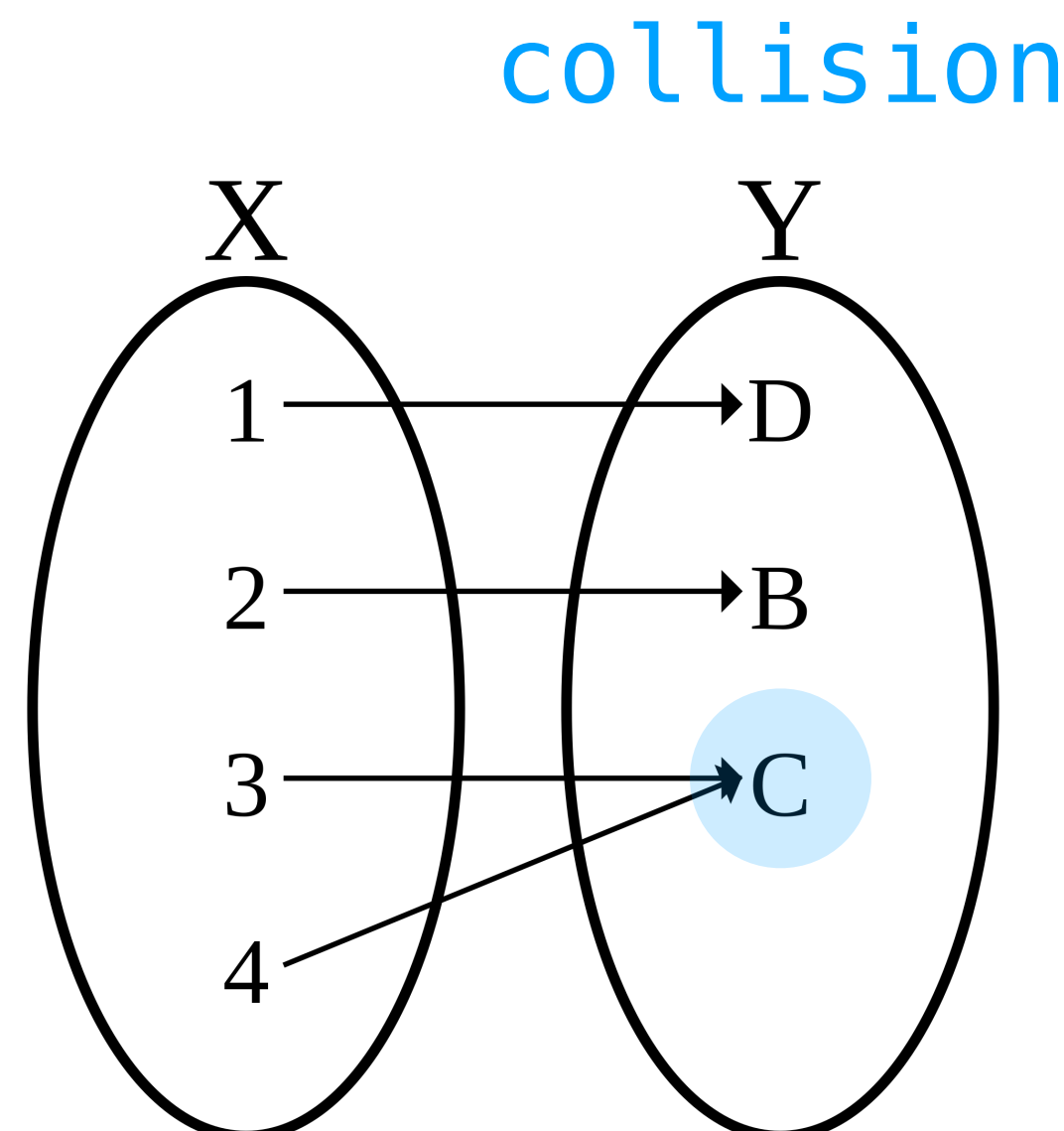
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Invertible transformations are 1-1 correspondences

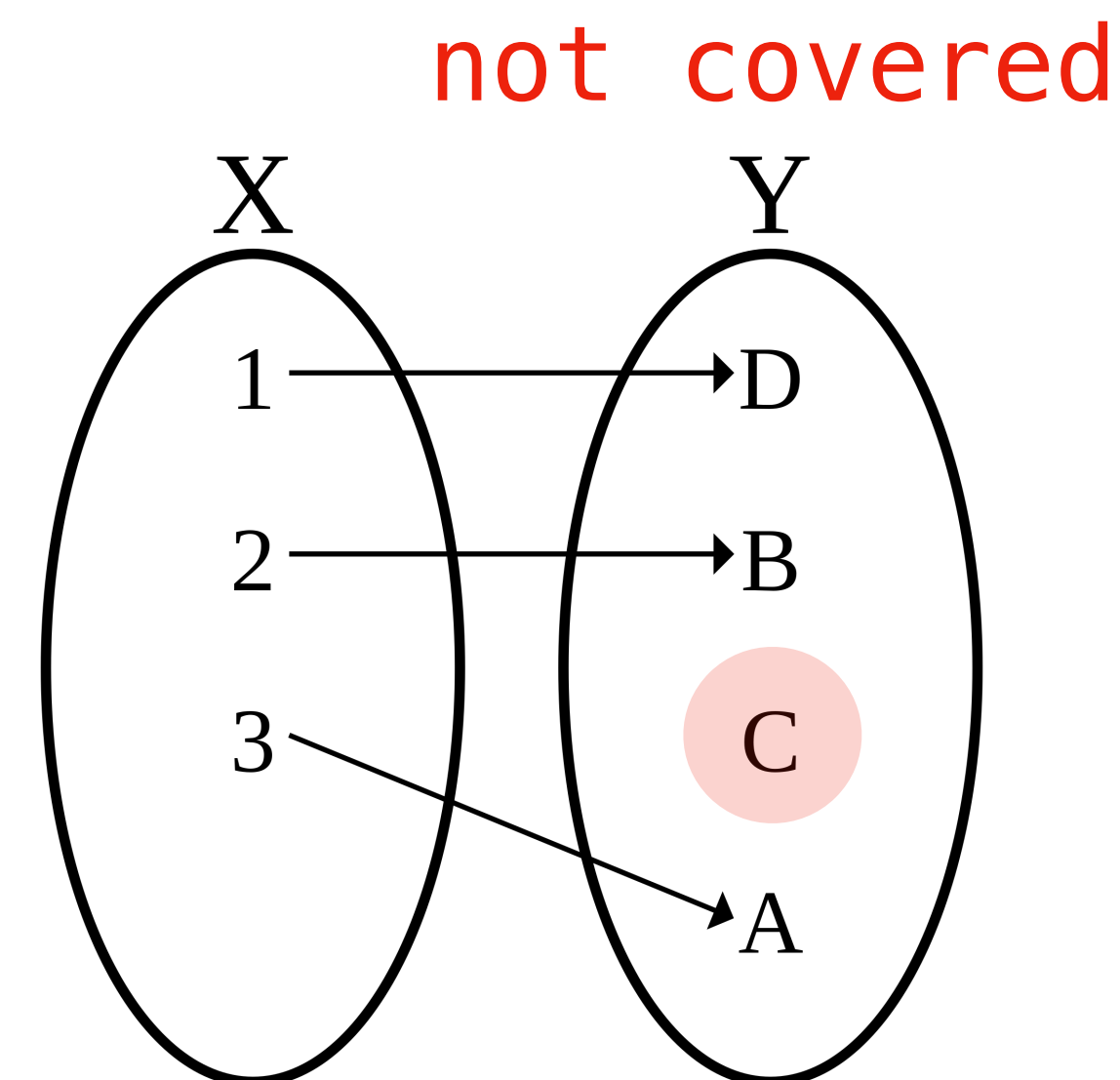
Kinds of Transformations (Pictorially)



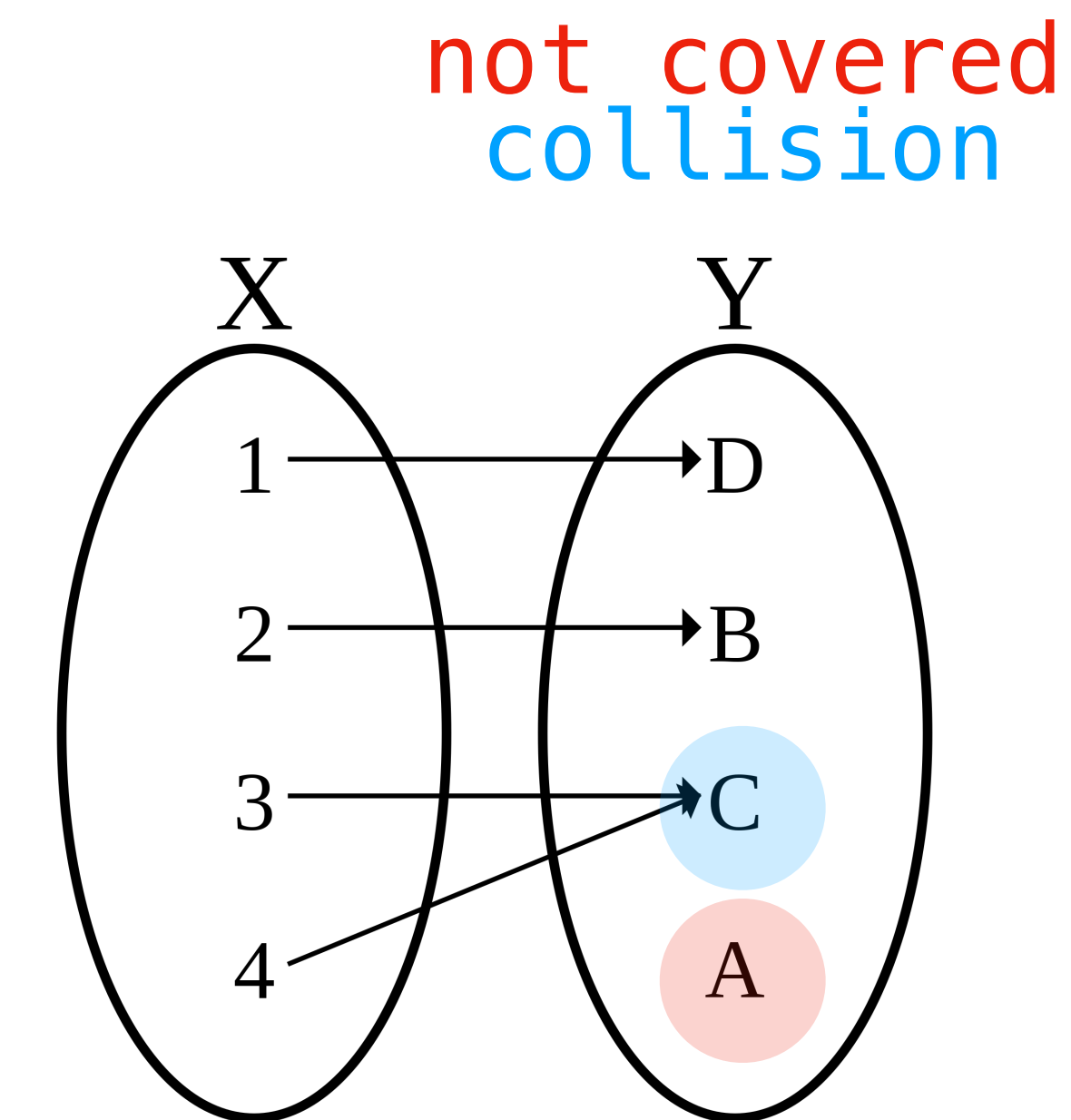
1-1 correspondence



onto, not 1-1



1-1 not onto



not 1-1, not onto

Computing Matrix Inverses

Fundamental Questions

How do we determine if A has an inverse?

How do we compute A^{-1} ?

In General

$$A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} = I$$

Can we solve for each $\mathbf{b}_i$?

In General

$$A\mathbf{b}_1 = \mathbf{e}_1$$

$$A\mathbf{b}_2 = \mathbf{e}_2$$

$$A\mathbf{b}_3 = \mathbf{e}_3$$

If we want a matrix B such that $AB = I$, then the above equation must hold (in the case B has 3 columns)

In General

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If we want a matrix B such that $AB = I$, then the above equation must hold (in the case B has 3 columns)

We need to solve 3 matrix equations

How To: Matrix Inverses

Q. Find the inverse of invertible $A \in \mathbb{R}^{n \times n}$

A. Solve the equation $A\mathbf{x} = \mathbf{e}_i$ for every standard basis vector $\mathbf{e}_i$. Put those solutions $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n$ into a single matrix

$$[\mathbf{s}_1 \quad \mathbf{s}_2 \quad \dots \quad \mathbf{s}_n]$$

How To: Matrix Inverses

Q. Find the inverse of invertible $A \in \mathbb{R}^{n \times n}$

A. Row reduce the matrix $[A \ I]$ to a matrix $[I \ B]$.
Then B is the inverse of A

This is really the same thing. It's a simultaneous reduction

demo

Special Case: 2×2 Matrice Inverses

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$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

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The inverse is defined only if the determinant is nonzero

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The **determinant** of a 2×2 matrix is the value $ad - bc$

The inverse is defined only if the determinant is nonzero

(see the notes on linear transformations for more information about determinants)

Example

$$\begin{bmatrix} -6 & 14 \\ 3 & -7 \end{bmatrix}$$

Algebra of Matrix Inverses

Challenge Problem

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Df. B is right inverse of A if $AB = I$

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Thm. If A has a right inverse B , then B is also a left inverse A

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Thm. If A has a right inverse B , then B is also a left inverse A

We only need to show A has a right/left inverse to show it is invertible

Algebraic Properties (Matrix Inverses)

Thm. For an invertible matrix A , the matrix A^{-1} is invertible

Algebraic Properties (Matrix Inverses)

Thm. For an invertible matrix A , the matrix A^T is invertible

Algebraic Properties (Matrix Inverses)

Thm. For a invertible matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times n}$, the matrix AB is invertible

Question

Suppose that A is a $n \times n$ invertible matrix such that $A = A^T$ and B is a $m \times n$ matrix

Simplify the expression $A(BA^{-1})^T$ using the algebraic properties we've seen

Answer: B^T

$$A(BA^{-1})^T$$

$$A = A^T$$

Invertible Matrix Theorem

The Point

Q. How do we know if a square matrix is invertible?

A. *Every* perspective we've taken so far can help us answer this question

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

1. A^T is invertible

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

- 2. $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$
- 3. $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b}$
- 4. $A\mathbf{x} = \mathbf{b}$ has at exactly one solution for every $\mathbf{b}$

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

5. A has a pivot in every column

6. A has a pivot in every row

7. A is row equivalent to I_n

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

- 8. The columns of A are linearly independent
- 9. The columns of A span $\mathbb{R}^n$
- 10. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

- 11. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto
- 12. $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one
- 13. $\mathbf{x} \mapsto A\mathbf{x}$ is a one-to-one correspondence
- 14. $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

Taking Stock: IMT

The following are logically equivalent:

1. A is invertible
2. A^T is invertible
3. $A\mathbf{x} = \mathbf{b}$ has at least one solution for any $\mathbf{b}$
4. $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
5. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$
6. A has n pivots (per row and per column)
7. A is row equivalent to I
8. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
9. The columns of A are linearly independent
10. The columns of A span $\mathbb{R}^n$
11. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto
12. $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one
13. $\mathbf{x} \mapsto A\mathbf{x}$ is a one-to-one correspondence
14. $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

These all express the
same thing

!! only for square matrices !!

We get a lot of information for free

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Thm. If A is square, then

A is 1-1 if and only if A is onto

We get a lot of information for free

Thm. If A is square, then

A **is 1-1** if and only if A **is onto**

We only need to check one of these

We get a lot of information for free

Thm. If A is square, then

A is 1-1 if and only if A is onto

We only need to check one of these

Warning. Remember this only applies square matrices

We get a lot of information for free

Theorem. If A is square, then

A is invertible $\equiv Ax = 0$ implies $x = 0$

Invertibility is completely determined by what A sends to 0

Question

If A is invertible, and B is row equivalent to A , is B also invertible?

Challenge Problem

If $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$ is invertible, is $[(\mathbf{a}_1 + \mathbf{a}_2 - 2\mathbf{a}_3) \ (\mathbf{a}_2 + 5\mathbf{a}_3) \ \mathbf{a}_3]$ also invertible?

Summary

The algebra of matrices can help us simplify matrix expressions

The invertible matrix theorem connects all the perspectives we've taken so far